

NAME

ioctl_ficlone, ioctl_ficlone – share some the data of one file with another file

LIBRARY

Standard C library (*libc*, *-lc*)

SYNOPSIS

```
#include <linux/fs.h>      /* Definition of FICLONE* constants */
#include <sys/ioctl.h>

int ioctl(int dest_fd, FICLONERANGE, struct file_clone_range *arg);
int ioctl(int dest_fd, FICLONE, int src_fd);
```

DESCRIPTION

If a filesystem supports files sharing physical storage between multiple files ("reflink"), this **ioctl(2)** operation can be used to make some of the data in the *src_fd* file appear in the *dest_fd* file by sharing the underlying storage, which is faster than making a separate physical copy of the data. Both files must reside within the same filesystem. If a file write should occur to a shared region, the filesystem must ensure that the changes remain private to the file being written. This behavior is commonly referred to as "copy on write".

This **ioctl** reflinks up to *src_length* bytes from file descriptor *src_fd* at offset *src_offset* into the file *dest_fd* at offset *dest_offset*, provided that both are files. If *src_length* is zero, the **ioctl** reflinks to the end of the source file. This information is conveyed in a structure of the following form:

```
struct file_clone_range {
    __s64 src_fd;
    __u64 src_offset;
    __u64 src_length;
    __u64 dest_offset;
};
```

Clones are atomic with regards to concurrent writes, so no locks need to be taken to obtain a consistent cloned copy.

The **FICLONE** **ioctl** clones entire files.

RETURN VALUE

On error, *-1* is returned, and *errno* is set to indicate the error.

ERRORS

Error codes can be one of, but are not limited to, the following:

EBADF

src_fd is not open for reading; *dest_fd* is not open for writing or is open for append-only writes; or the filesystem which *src_fd* resides on does not support reflink.

EINVAL

The filesystem does not support reflinking the ranges of the given files. This error can also appear if either file descriptor represents a device, FIFO, or socket. Disk filesystems generally require the offset and length arguments to be aligned to the fundamental block size. XFS and Btrfs do not support overlapping reflink ranges in the same file.

EISDIR

One of the files is a directory and the filesystem does not support shared regions in directories.

EOPNOTSUPP

This can appear if the filesystem does not support reflinking either file descriptor, or if either file descriptor refers to special inodes.

EPERM

dest_fd is immutable.

ETXTBSY

One of the files is a swap file. Swap files cannot share storage.

EXDEV

dest_fd and *src_fd* are not on the same mounted filesystem.

STANDARDS

Linux.

HISTORY

Linux 4.5.

They were previously known as **BTRFS_IOC_CLONE** and **BTRFS_IOC_CLONE_RANGE**, and were private to Btrfs.

NOTES

Because a copy-on-write operation requires the allocation of new storage, the **fallocate(2)** operation may unshare shared blocks to guarantee that subsequent writes will not fail because of lack of disk space.

SEE ALSO

ioctl(2)